1) Given, $k=3$

$P_1 = C = 20\,kN$

$L_1 = 10^6$ rev

a) When $P_2 = 10\,kN$,

$$L_D = 10^6\left(\frac{20}{10}\right)^3$$

$$= 8\times10^6 \text{ rev}$$

b) When $P_2 = 25\,kN$,

$$L_D = 10^6\left(\frac{20}{25}\right)^3$$

$$= 5.12\times10^5 \text{ rev}$$

2) $F_r = 1.5\,kN$

$F_a = 0.9\,kN$

$P = XVF_r + YF_a$

$= XVF_r + YF_a$

Assume $Y = 1.5$,

From table 14-5:

$$\frac{F_a}{F_r} = \frac{0.9}{1.5} = 0.6$$

$X = 0.56$

$P = 0.56(1.0)(1.5) + 1.5(0.9)$

$= 2.19\,kN$

$$C = P\left(\frac{Ld}{10^6}\right)^{\frac{1}{k}} = 2.19\left(\frac{720 \times 60 \times 20000}{10^6}\right)^{\frac{1}{3}}$$

$= 20.858\,kN$

From table 14-3, select bearing 6305

$\because C_{selected} > C_{calculated}$

$22.50 > 20.858$

$C_0 = 11.6\,kN$

Find $\frac{F_a}{C_o}$:

$$\frac{F_a}{C_o} = \frac{0.9}{11.6} = 0.0776$$

From table 14-5, determing e:

$$\frac{e - 0.27}{0.28 - 0.27} = \frac{0.0776 - 0.070}{0.084 - 0.070}$$

$$e = 0.275$$

Check if $\frac{F_a}{F_r} > e$:

$$\frac{F_a}{F_r} = \frac{0.9}{1.5} = 0.6 > 0.275$$

Finding the new thrust factor:

$$\frac{Y' - 1.55}{1.63 - 1.55} = \frac{0.0766 - 0.070}{0.084 - 0.070}$$

$$Y' = 1.59$$

New equivalent load:

$$P' = XVF_r + Y'F_a$$
$$= 0.56(1)(1.5) + 1.59(0.9)$$
$$= 2.271 \text{kN}$$

2) New dynamic load rating:

$$C' = P'\left(\frac{L_d}{10^6}\right)^{\frac{1}{k}} = 2.271\left(\frac{720 \times 60 \times 20000}{10^6}\right)^{\frac{1}{3}}$$

$$= 21.63\,kN$$

Since $21.63 < 22.50\,kN$,
bearing 6305 is suitable.

3) $F_r = 2.8\,kN$

$F_a = 0.9\,kN$

Let $Y = 1.5$,

$$P = XVF_r + YF_a$$
$$= 0.56(1)(2.8) + 1.5(0.9)$$
$$= 2.918\,kN$$

$$C = P\left(\frac{L_d}{10^6}\right)^{\frac{1}{k}}$$

$$\left(\frac{L_d}{10^6}\right)^{\frac{1}{k}} = \left(\frac{720 \times 60 \times 20000}{10^6}\right)^{\frac{1}{3}}$$
$$= 9.52$$

$$C = 2.918(9.52)$$
$$= 27.79\,kN$$

From table 14-3, bearing 6306 is selected

$\because C_{selected} > C_{calculated}$

$28.1 > 27.79$

$$C_0 = 16.0\,kN$$

$$\frac{F_a}{C_0} = \frac{0.9}{16} = 0.05625$$

3) From table 14-5:

$$\frac{e-0.26}{0.27-0.26} = \frac{0.05625-0.056}{0.070-0.056}$$

$$e = 0.2602$$

$$\frac{F_a}{F_r} = \frac{0.9}{2.8} = 0.3214 > 0.2602$$

New thrust factor:

$$\frac{Y'-1.63}{1.71-1.63} = \frac{0.05625-0.056}{0.070-0.056}$$

$$Y' = 1.63$$

$$P' = XVF_r + Y'F_a$$

$$= 0.56(1)(2.8) + 1.63(0.9)$$

$$= 3.035\,kN$$

$$C' = P'\left(\frac{L_d}{10^6}\right)^{\frac{1}{k}}$$

$$= 3.035(9.52)$$

$$= 28.907\,kN$$

Since $C' > 28.1\,kN$, bearing 6306 is unsuitable, so selecting bearing 6307 as $C_{selected} = 33.2\,kN > 28.907\,kN$